

9.2.2 Smoothing constant states

Now we will think about the improvement (due to smoothing) in the estimation accuracy of constant states. If the system states are constant then $F_k = I$ and $Q = 0$. Equation (9.25) shows that

$$\begin{aligned} P_{k+1} &= F_k P_k (F_k - L_k H_k)^T + Q_k \\ &= P_k (I - L_k H_k)^T \\ \Sigma_{k+1} &= \Sigma_k (F_k - L_k H_k)^T \\ &= \Sigma_k (I - L_k H_k)^T \end{aligned} \quad (9.36)$$

Comparing these expressions for P_{k+1} and Σ_{k+1} , and realizing from Equation (9.24) that the initial value of $\Sigma_j = P_j$, we see that $\Sigma_k = P_k$ for $k \geq j$. This means that the expression for L_k from Equation (9.25) can be written as

$$\begin{aligned} L_k &= F_k P_k H_k^T (H_k P_k H_k^T + R_k)^{-1} \\ &= \Sigma_k H_k^T (H_k P_k H_k^T + R_k)^{-1} \\ &= \lambda_k \end{aligned} \quad (9.37)$$

Substituting these results into the expression for Π_{k+1} from Equation (9.25) we see that

$$\begin{aligned} \Pi_{k+1} &= \Pi_k - \Sigma_k H_k^T \lambda_k^T \\ &= \Pi_k - P_k H_k^T L_k^T \end{aligned} \quad (9.38)$$

Realizing that the initial value of $\Pi_j = P_j$, and comparing this expression for Π_{k+1} with Equation (9.36) for P_{k+1} , we see that $\Pi_k = P_k$ for $k \geq j$. Recall that P_k is the covariance of the estimate of x_k from the standard Kalman filter, and Π_k is the covariance of the estimate of x_j given measurements up to and including time $(k - 1)$.

This result shows that constant states are not smoothable. Additional measurements are still helpful for refining an estimate of a constant state. However, there is no point to using smoothing for estimation of a constant state. If we want to estimate a constant state at time j using measurements up to time $k > j$, then we may as well simply run the standard Kalman filter up to time k . Implementing the smoothing equations will not gain any improvement in estimation accuracy.

9.3 FIXED-LAG SMOOTHING

In fixed-lag smoothing we want to obtain an estimate of the state at time $(k - N)$ given measurements up to and including time k , where the time index k continually changes as we obtain new measurements, but the lag N is a constant. In other words, at each time point we have N future measurements available for our state estimate. We therefore want to obtain $\hat{x}_{k-N,k}$ for $k = N, N + 1, \dots$, where N is a fixed positive integer. This could be the case, for example, if a satellite is continually taking photographs that are to be displayed or transmitted N time steps after the photograph is taken. In this case, since the photograph is processed N time steps after it is taken, we have N additional measurements after each photograph that

are available to update the estimate of the satellite state and hence improve the quality of the photograph. In this section we use the notation

$$\begin{aligned}\hat{x}_{k-N,k} &= E(x_{k-N}|y_1, \dots, y_k) \\ \Pi_{k-N} &= E[(x_{k-N} - \hat{x}_{k-N,k})(x_{k-N} - \hat{x}_{k-N,k})^T]\end{aligned}\quad (9.39)$$

Note that the notation has changed slightly from the previous section. In the previous section we used the notation $\hat{x}_{k,m}$ to refer to the estimate of x_k given measurements up to and including time $(m-1)$. In this section (and in the remainder of this chapter) we use $\hat{x}_{k,m}$ to refer to the estimate of x_k given measurements up to and including time m .

Let us define $x_{k,m}$ as the state x_{k-m} propagated with an identity transition matrix and zero process noise to time k . With this definition we see that

$$\begin{aligned}x_{k+1,1} &= x_k \\ x_{k+1,2} &= x_{k-1} \\ &= x_{k,1} \\ x_{k+1,3} &= x_{k-2} \\ &= x_{k,2} \\ &\text{etc.}\end{aligned}\quad (9.40)$$

We can therefore define the augmented system

$$\begin{aligned}\begin{bmatrix} x_{k+1} \\ x_{k+1,1} \\ \vdots \\ x_{k+1,N+1} \end{bmatrix} &= \begin{bmatrix} F_k & 0 & \cdots & 0 \\ I & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & I & 0 \end{bmatrix} \begin{bmatrix} x_k \\ x_{k,1} \\ \vdots \\ x_{k,N+1} \end{bmatrix} + \begin{bmatrix} I \\ 0 \\ \vdots \\ 0 \end{bmatrix} w_k \\ y_k &= [H_k \ 0 \ \cdots \ 0] \begin{bmatrix} x_k \\ x_{k,1} \\ \vdots \\ x_{k,N+1} \end{bmatrix} + v_k\end{aligned}\quad (9.41)$$

The Kalman filter estimates of the components of this augmented state vector are given as

$$\begin{aligned}E(x_{k+1}|y_1 \cdots y_k) &= \hat{x}_{k+1}^- \\ &= \hat{x}_{k+1,k} \\ E(x_{k+1,1}|y_1 \cdots y_k) &= E(x_k|y_1 \cdots y_k) \\ &= \hat{x}_k^+ \\ &= \hat{x}_{k,k} \\ E(x_{k+1,2}|y_1 \cdots y_k) &= E(x_{k-1}|y_1 \cdots y_k) \\ &= \hat{x}_{k-1,k} \\ &\vdots \\ E(x_{k+1,N+1}|y_1 \cdots y_k) &= \hat{x}_{k-N,k}\end{aligned}\quad (9.42)$$

We see that if we can use a Kalman filter to estimate the states of the augmented system (using measurements up to and including time k), then the estimate of the

last element of the augmented state vector, $x_{k+1,N+1}$, will be equal to the estimate of x_{k-N} given measurements up to and including time k . This is the estimate that we are looking for in fixed-lag smoothing. This idea is illustrated in Figure 9.7.

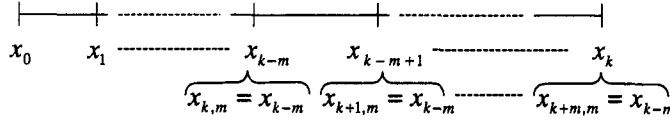


Figure 9.7 This illustrates the idea that is used to obtain the fixed-lag smoother. A fictitious state variable $x_{k,m}$ is initialized as $x_{k,m} = x_{k-m}$ and from that point on has an identity state transition matrix. The *a posteriori* estimate of $x_{k+m,m}$ is then equal to $\hat{x}_{k-m,k}$.

From Equation (9.10) we can write the Kalman filter for the augmented system of Equation (9.41) as follows:

$$\begin{bmatrix} \hat{x}_{k+1}^- \\ \hat{x}_{k,k} \\ \vdots \\ \hat{x}_{k-N,k} \end{bmatrix} = \begin{bmatrix} F_k & 0 & \cdots & 0 \\ I & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & I & 0 \end{bmatrix} \begin{bmatrix} \hat{x}_k^- \\ \hat{x}_{k-1,k-1} \\ \vdots \\ \hat{x}_{k-(N+1),k-1} \end{bmatrix} + \begin{bmatrix} L_{k,0} \\ L_{k,1} \\ \vdots \\ L_{k,N+1} \end{bmatrix} \left(y_k - \begin{bmatrix} H_k & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} \hat{x}_k^- \\ \hat{x}_{k-1,k-1} \\ \vdots \\ x_{k-(N+1),k-1} \end{bmatrix} \right) \quad (9.43)$$

where the $L_{k,i}$ matrices are components of the smoother gain that will be determined in this section. Note that $L_{k,0}$ is the standard Kalman gain. The smoother gain L_k is defined as

$$L_k = \begin{bmatrix} L_{k,0} \\ L_{k,1} \\ \vdots \\ L_{k,N+1} \end{bmatrix} \quad (9.44)$$

From Equation (9.10) we see that the L_k gain matrix is given by

$$L_k = \begin{bmatrix} F_k & 0 & \cdots & 0 \\ I & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & I & 0 \end{bmatrix} \begin{bmatrix} P_k^{0,0} & \cdots & (P_k^{0,N+1})^T \\ \vdots & \ddots & \vdots \\ P_k^{0,N+1} & \cdots & P_k^{N+1,N+1} \end{bmatrix} \begin{bmatrix} H_k^T \\ 0 \\ \vdots \\ 0 \end{bmatrix} \times \left(\begin{bmatrix} H_k & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} P_k^{0,0} & \cdots & (P_k^{0,N+1})^T \\ \vdots & \ddots & \vdots \\ P_k^{0,N+1} & \cdots & P_k^{N+1,N+1} \end{bmatrix} \begin{bmatrix} H_k^T \\ 0 \\ \vdots \\ 0 \end{bmatrix} + R_k \right)^{-1} \quad (9.45)$$

where the $P_k^{i,j}$ covariance matrices are defined as

$$P_k^{i,j} = E[(x_{k-j} - \hat{x}_{k-j,k-1})(x_{k-i} - \hat{x}_{k-i,k-1})^T] \quad (9.46)$$

The L_k expression above can be simplified to

$$L_k = \begin{bmatrix} F_k P_k^{0,0} H_k^T \\ P_k^{0,0} H_k^T \\ \vdots \\ P_k^{0,N} H_k^T \end{bmatrix} (H_k P_k^{0,0} H_k^T + R_k)^{-1} \quad (9.47)$$

From Equation (9.10) we see that the covariance-update equation for the Kalman filter for our augmented system can be written as

$$\begin{bmatrix} P_{k+1}^{0,0} & \cdots & (P_{k+1}^{0,N+1})^T \\ \vdots & \ddots & \vdots \\ P_{k+1}^{0,N+1} & \cdots & P_{k+1}^{N+1,N+1} \end{bmatrix} = \begin{bmatrix} F_k & 0 & \cdots & 0 \\ I & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & I & 0 \end{bmatrix} \begin{bmatrix} P_k^{0,0} & \cdots & (P_k^{0,N+1})^T \\ \vdots & \ddots & \vdots \\ P_k^{0,N+1} & \cdots & P_k^{N+1,N+1} \end{bmatrix} \times \\ \left(\begin{bmatrix} F_k^T & I & \cdots & 0 \\ 0 & 0 & \ddots & \vdots \\ \vdots & \ddots & \ddots & I \\ 0 & \cdots & \cdots & 0 \end{bmatrix} - \begin{bmatrix} H_k^T \\ 0 \\ \vdots \\ 0 \end{bmatrix} L_k^T \right) + \begin{bmatrix} Q_k & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & \cdots & 0 \end{bmatrix} \quad (9.48)$$

Substituting for L_k from Equation (9.47) and multiplying out gives

$$\begin{bmatrix} P_{k+1}^{0,0} & \cdots & (P_{k+1}^{0,N+1})^T \\ \vdots & \ddots & \vdots \\ P_{k+1}^{0,N+1} & \cdots & P_{k+1}^{N+1,N+1} \end{bmatrix} = \begin{bmatrix} F_k P_k^{0,0} & F_k (P_k^{0,1})^T & \cdots & F_k (P_k^{0,N+1})^T \\ P_k^{0,0} & (P_k^{0,1})^T & \cdots & (P_k^{0,N+1})^T \\ \vdots & \ddots & \ddots & \vdots \\ P_k^{0,N} & P_k^{1,N} & \cdots & (P_k^{N,N+1})^T \end{bmatrix} \times \\ \left(\begin{bmatrix} F_k^T & I & \cdots & 0 \\ 0 & 0 & \ddots & \vdots \\ \vdots & \ddots & \ddots & I \\ 0 & \cdots & \cdots & 0 \end{bmatrix} - H_k^T (H_k P_k^{0,0} H_k^T + R_k)^{-1} H_k \times \right. \\ \left. \begin{bmatrix} P_k^{0,0} F_k^T & P_k^{0,0} & \cdots & P_k^{0,N} \\ 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix} \right) + \begin{bmatrix} Q_k & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & \cdots & 0 \end{bmatrix} \quad (9.49)$$

This gives us the update equations for the P matrices. The equations for the first column of the P matrix are as follows:

$$\begin{aligned} P_{k+1}^{0,0} &= F_k P_k^{0,0} \left[F_k^T - H_k^T (H_k P_k^{0,0} H_k^T + R_k)^{-1} H_k P_k^{0,0} F_k^T \right] + Q_k \\ &= F_k P_k^{0,0} (F_k - L_{k,0} H_k)^T + Q_k \\ P_{k+1}^{0,1} &= P_k^{0,0} (F_k - L_{k,0} H_k)^T \\ &\vdots \\ P_{k+1}^{0,N+1} &= P_k^{0,N} (F_k - L_{k,0} H_k)^T \end{aligned} \quad (9.50)$$

The equations for the diagonal elements of the P matrix are as follows:

$$\begin{aligned}
 P_{k+1}^{1,1} &= P_k^{0,0} \left[I - H_k^T (H_k P_k^{0,0} H_k^T + R_k)^{-1} H_k P_k^{0,0} F_k^T \right] \\
 &= P_k^{0,0} - P_k^{0,0} H_k^T L_{k,1}^T F_k^T \\
 P_{k+1}^{2,2} &= P_k^{0,1} [-H_k^T L_{k,2}^T F_k^T] + P_k^{1,1} \\
 &= P_k^{1,1} - P_k^{0,1} H_k^T L_{k,2}^T F_k^T \\
 &\vdots \\
 P_{k+1}^{i,i} &= P_k^{i-1,i-1} - P_k^{0,i-1} H_k^T L_{k,i}^T F_k^T
 \end{aligned} \tag{9.51}$$

These equations give us the formulas that we can use for fixed-lag smoothing. This gives us the estimate $E(x_{k-N}|y_1, \dots, y_k)$ for a fixed N as k continually increments. The fixed-lag smoother is summarized as follows.

The fixed-lag smoother

1. Run the standard Kalman filter of Equation (9.10) to obtain $\hat{x}_{k+1}^-$, L_k , and P_k^- .
2. Initialize the fixed-lag smoother as follows:

$$\begin{aligned}
 \hat{x}_{k+1,k} &= \hat{x}_{k+1}^- \\
 L_{k,0} &= L_k \\
 P_k^{0,0} &= P_k^-
 \end{aligned} \tag{9.52}$$

3. For $i = 1, \dots, N+1$, perform the following:

$$\begin{aligned}
 L_{k,i} &= P_k^{0,i-1} H_k^T (H_k P_k^{0,0} H_k^T + R_k)^{-1} \\
 P_{k+1}^{i,i} &= P_k^{i-1,i-1} - P_k^{0,i-1} H_k^T L_{k,i}^T F_k^T \\
 P_{k+1}^{0,i} &= P_k^{0,i-1} (F_k - L_{k,0} H_k)^T \\
 \hat{x}_{k+1-i,k} &= \hat{x}_{k+2-i,k} + L_{k,i} (y_k - H_k \hat{x}_k^-)
 \end{aligned} \tag{9.53}$$

Note that the first time through this loop is the measurement update of the standard Kalman filter. At the end of this loop we have the smoothed estimates of *each* state with delays between 0 and N , given measurements up to and including time k . These estimates are denoted $\hat{x}_{k,k}, \dots, \hat{x}_{k-N,k}$. We also have the estimation-error covariances, denoted $P_{k+1}^{1,1}, \dots, P_{k+1}^{N+1,N+1}$.

The percent improvement due to smoothing can be computed as

$$\text{Percent Improvement} = \frac{100 \operatorname{Tr}(P_k^{0,0} - P_k^{N+1,N+1})}{\operatorname{Tr}(P_k^{0,0})} \tag{9.54}$$